APPLICATION LAYER

Chapter 6

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DOMAIN NAME SYSTEM

- *DNS* translates *domain names* to IP addresses so browsers can load Internet resources.
- Each device connected to the Internet has a unique IP address which other machines use to find the device.
- *DNS servers* eliminate the need for humans to memorize IP addresses such as 192.168.45.23

Introduction

1. What is the IP address of udel.edu ?

It is 128.175.13.92

1. What is the host name of 128.175.13.74

It is strauss.udel.edu

DOMAIN NAME SYSTEM

- The **Domain Name System** (**DNS**) is a hierarchical and decentralized naming system for computers, services, or other resources connected to the Internet or a private network.
- It associates various information with domain names assigned to each of the participating entities.
- It translates more readily memorized domain names to the numerical IP addresses needed for locating and identifying computer services and devices with the underlying network protocols.

DNS COMPONENTS

There are 3 components:

- Name Space:

  Specifications for a structured unique names and data associated with the names

- Resolvers:

  Client programs that extract information from Name Servers.

- Name Servers:

  Server programs which hold information about the structure and the names.

Domain Name Space

Flat Name Space

A name in this space is a sequence of characters without structure.

Network Information System (NIS),a central site looked after administration of name space.

The main disadvantage of a flat name space is that it cannot be used in a large system such as the Internet because it must be centrally controlled to avoid ambiguity and duplication.

Domain Name Space

Hierarchical Name Space

In a hierarchical name space, each name is made of several parts. The first part can define the nature of the organization, the second part can define the name of an organization, the third part can define departments in the organization, and so on.

In this case, the authority to assign and control the name spaces can be decentralized.

A central authority can assign the part of the name that defines the nature of the organization and the name of the organization. The responsibility of the rest of the name can be given to the organization itself.

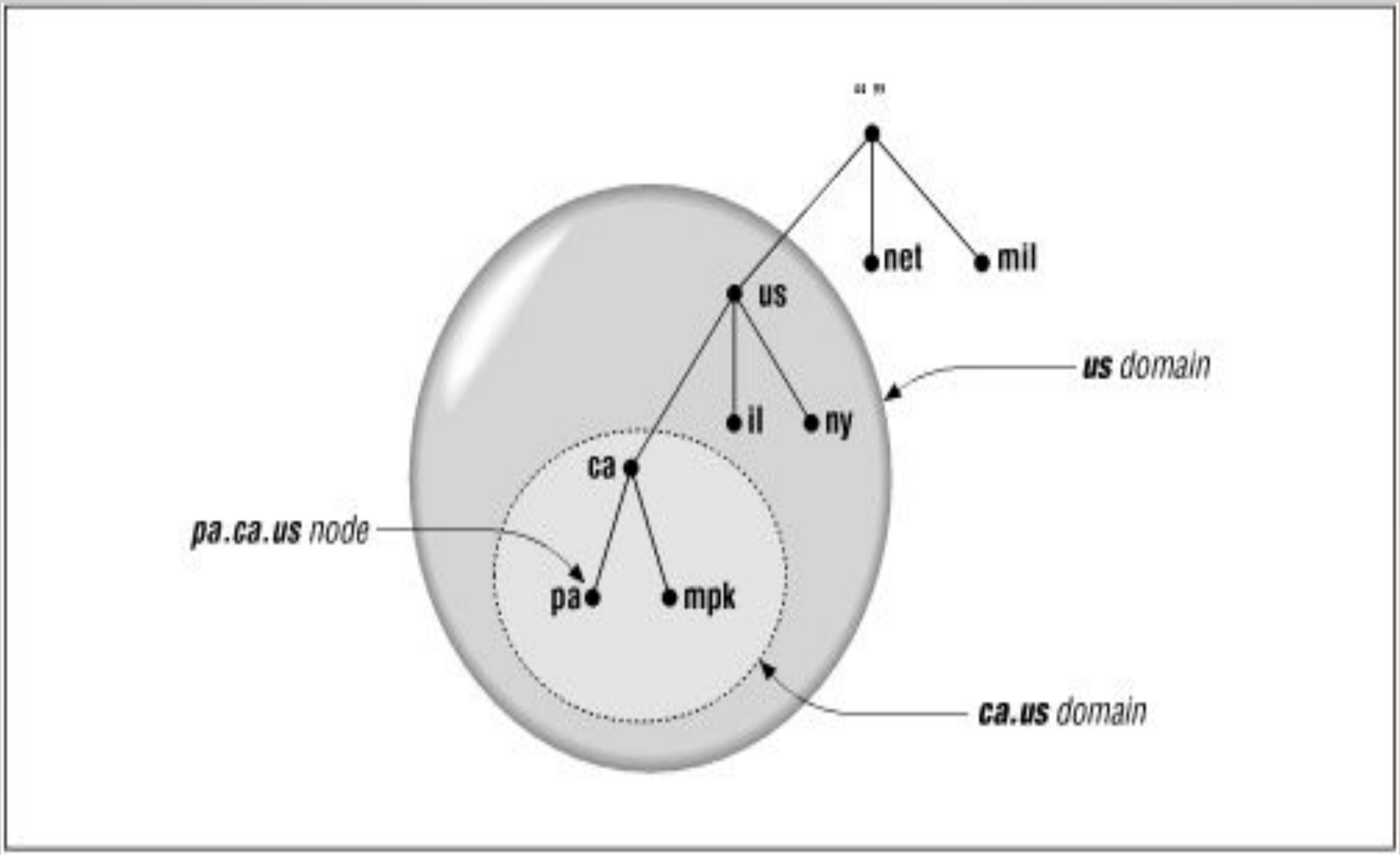
Domain Name Space

The mechanism that implements hierarchical name space is called **Domain Name Space**.
In this design the names are defined in an inverted-tree structure with the root at the top. The tree can have only 128 levels: level 0 (root) to level 127.

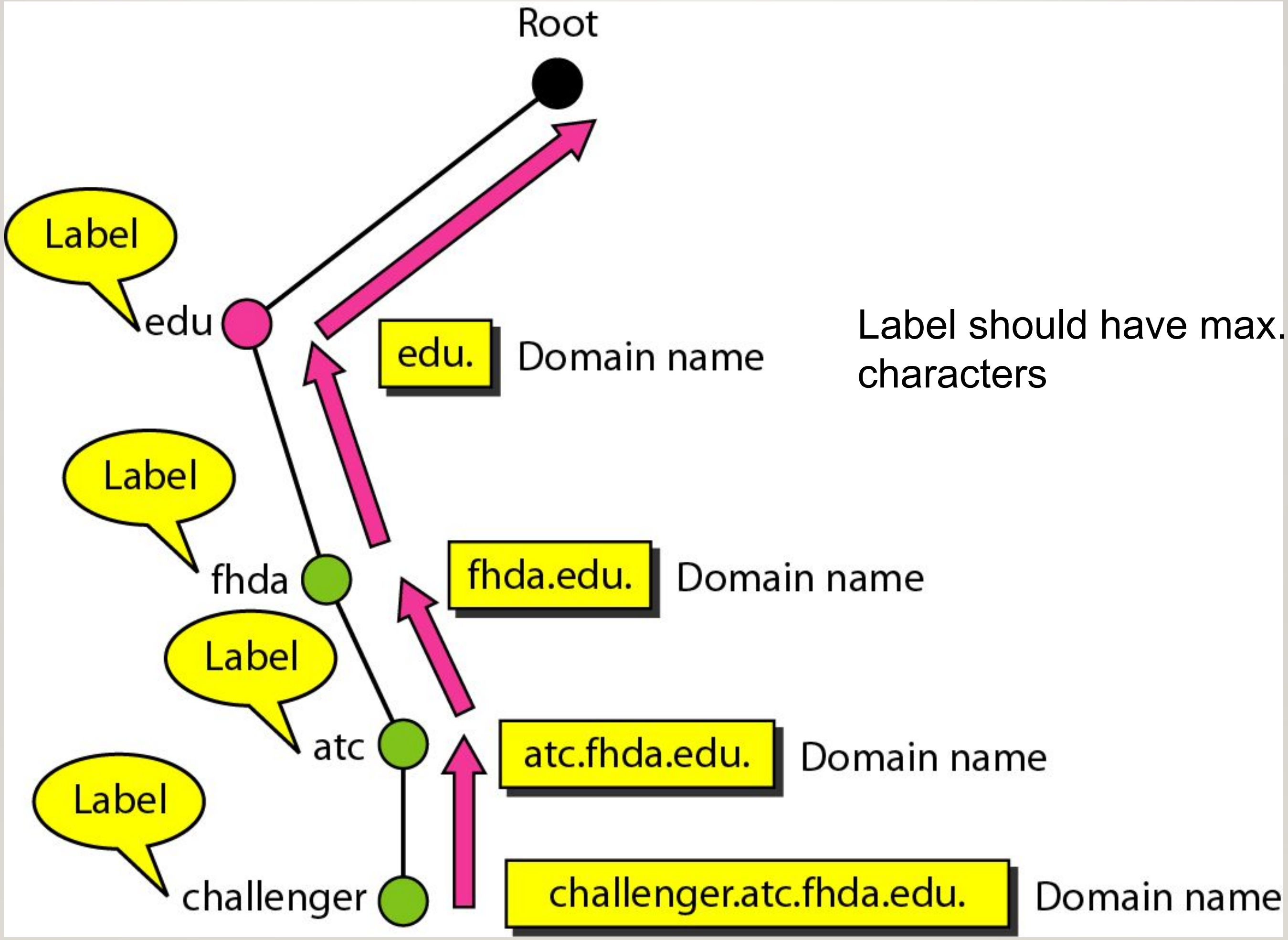
Topics discussed in this section:

Label
Domain Name
Domain

Domain



Domain names and labels



Label should have max. 63 characters

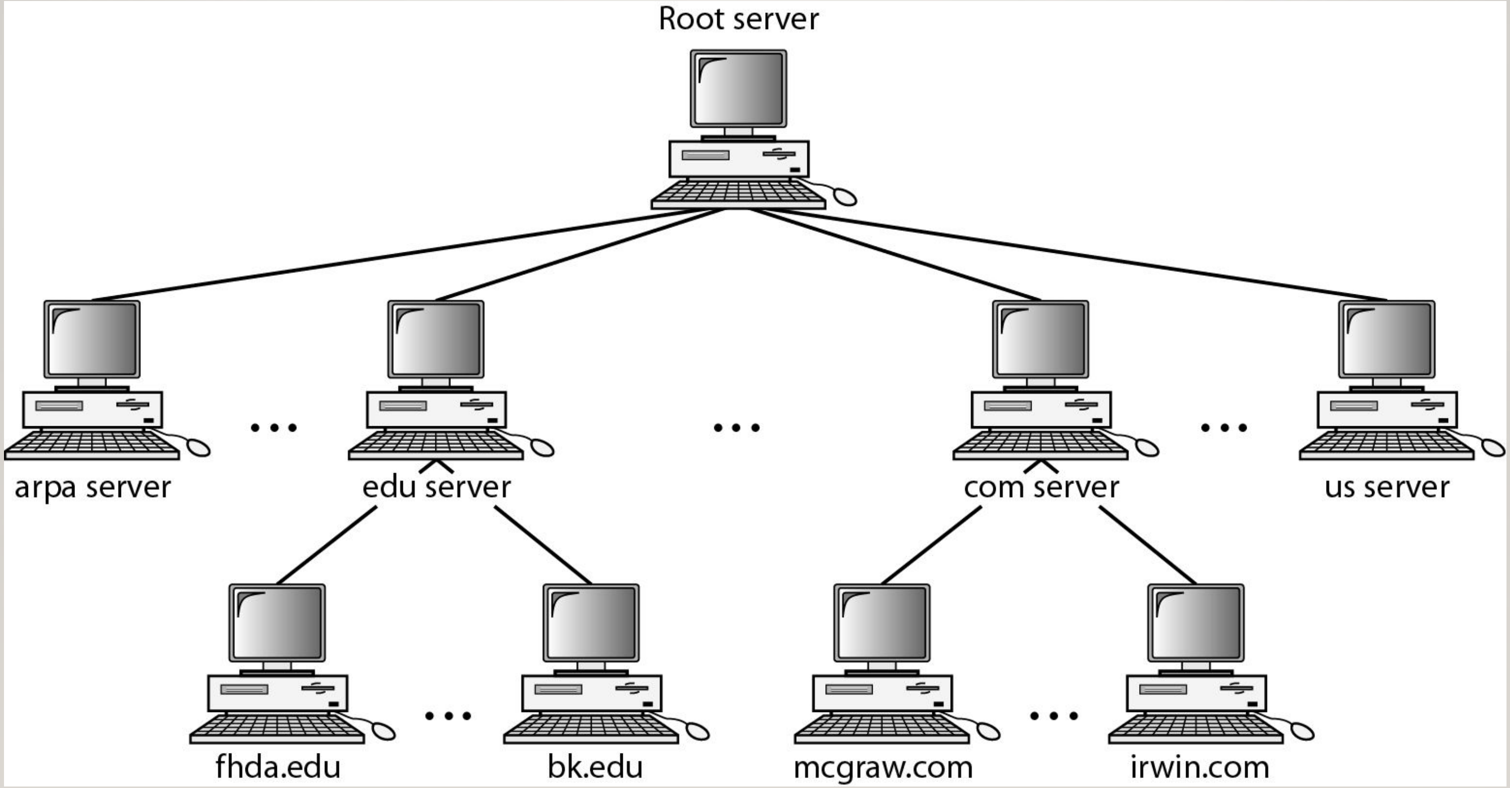
DISTRIBUTION OF NAME SPACE

The information contained in the domain name space must be stored. However, it is very inefficient and also unreliable to have just one computer store such a huge amount of information. In this section, we discuss the distribution of the domain name space.

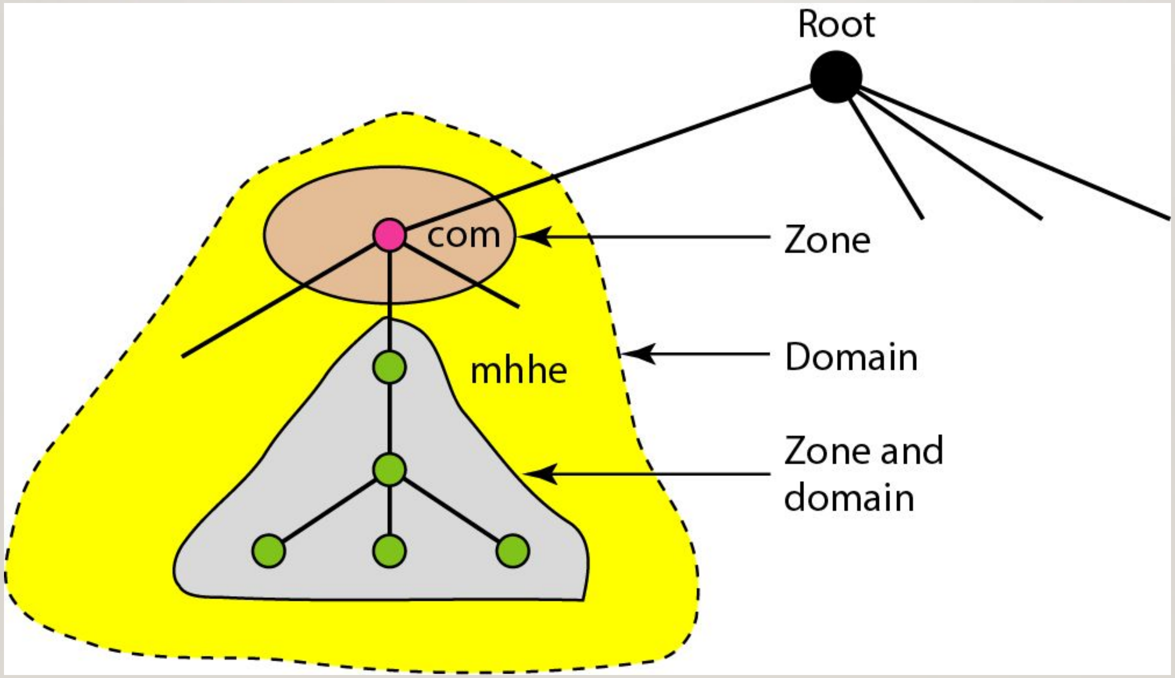
Topics discussed in this section:

Hierarchy of Name Servers
Zone
Root Server
Primary and Secondary Servers

Hierarchy of name servers



Zones and domains



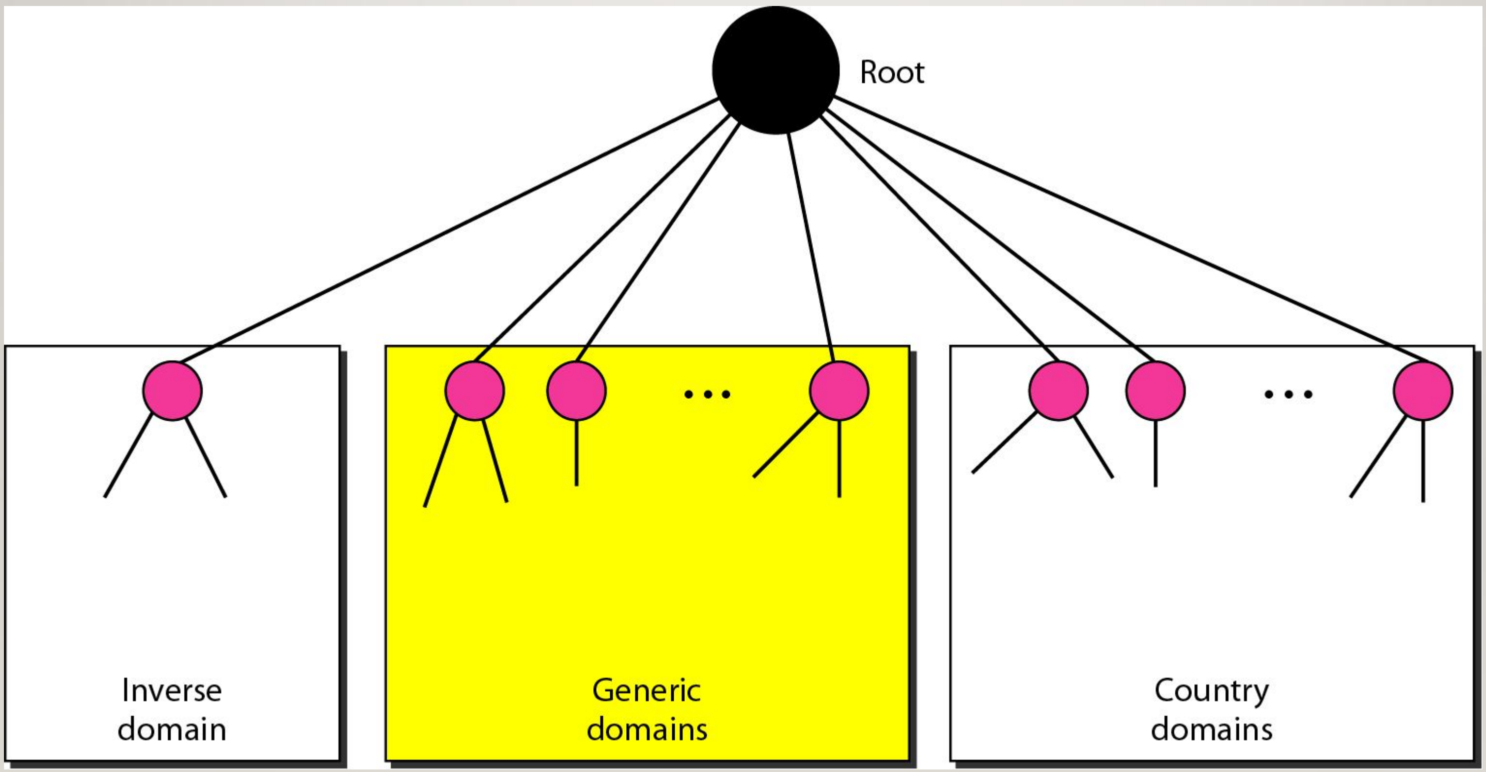
DNS IN THE INTERNET

DNS is a protocol that can be used in different platforms. In the Internet, the domain name space (tree) is divided into three different sections: generic domains, country domains, and the inverse domain.

Topics discussed in this section:

Generic Domains
Country Domains
Inverse Domain

DNS IN THE INTERNET



Generic domain labels

Label	*Description*
aero	Airlines and aerospace companies
biz	Businesses or firms (similar to "com")
com	Commercial organizations
coop	Cooperative business organizations
edu	Educational institutions
gov	Government institutions
info	Information service providers
int	International organizations
mil	Military groups
museum	Museums and other nonprofit organizations
name	Personal names (individuals)
net	Network support centers
org	Nonprofit organizations
pro	Professional individual organizations

RESOLUTION

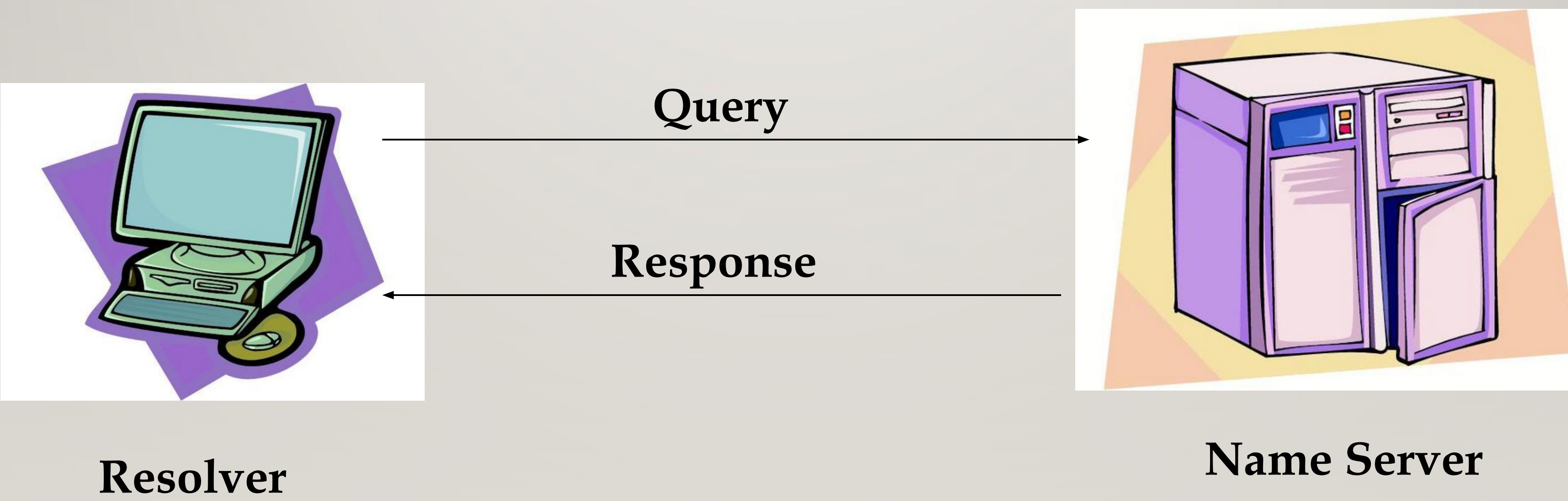
Mapping a name to an address or an address to a name is called name-address resolution.

Topics discussed in this section:

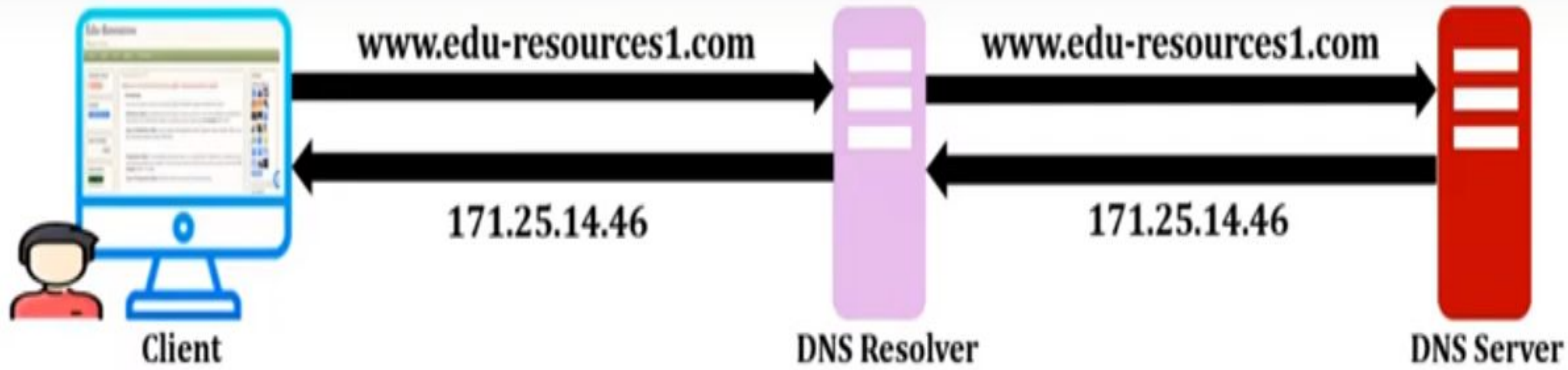
Resolver
Mapping Names to Addresses
Mapping Addresses to Names
Recursive Resolution
Iterative Resolution
Caching

DNS RESOLVERS

A Resolver maps a name to an address and vice versa.



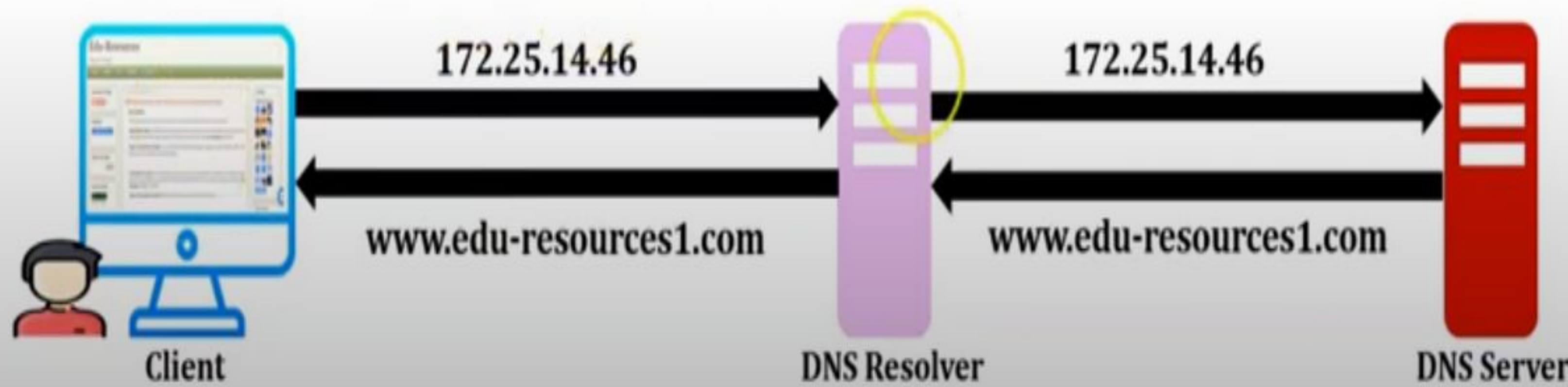
- Mapping a domain name to an IP address



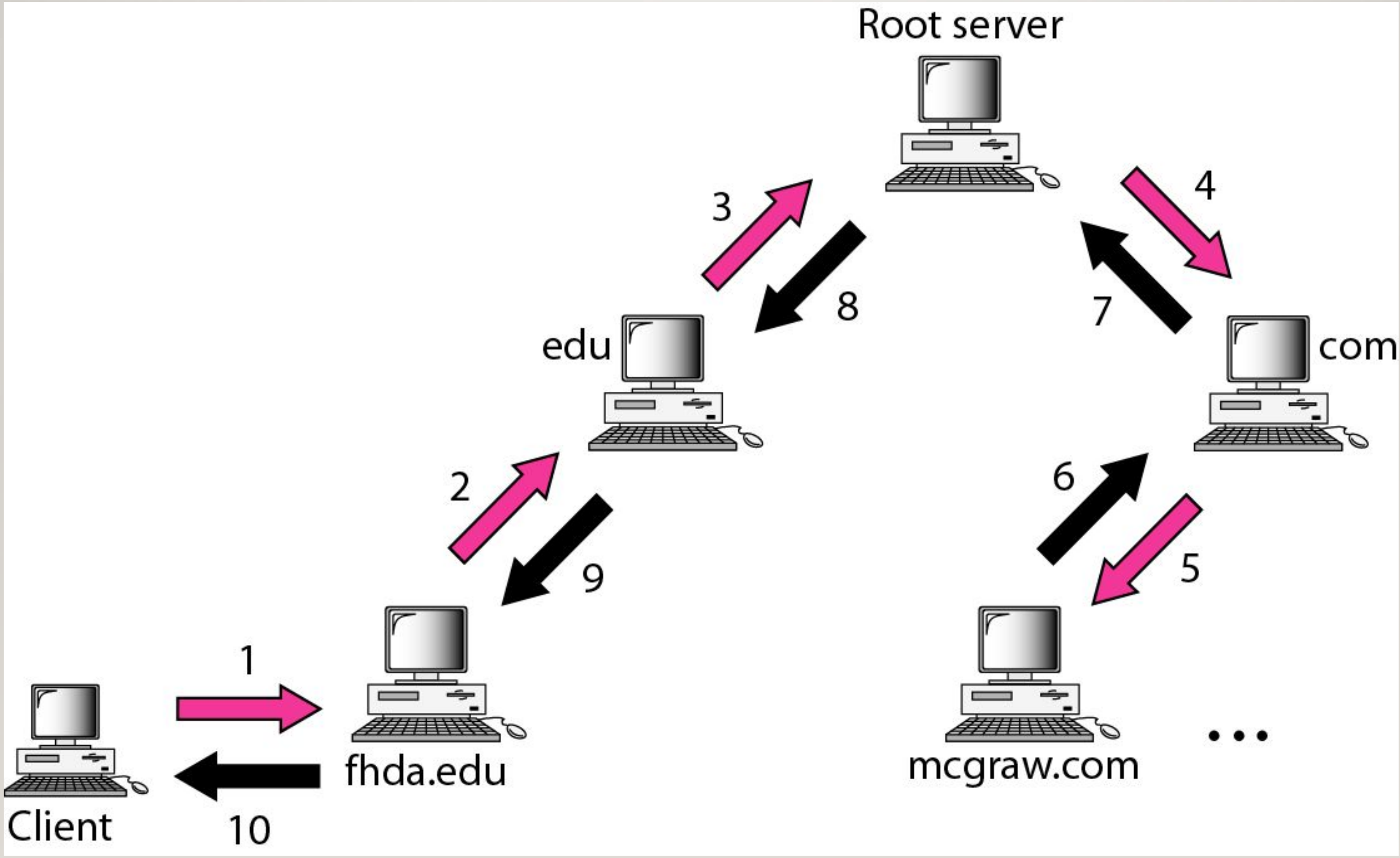
Generic Domain

Country Domain

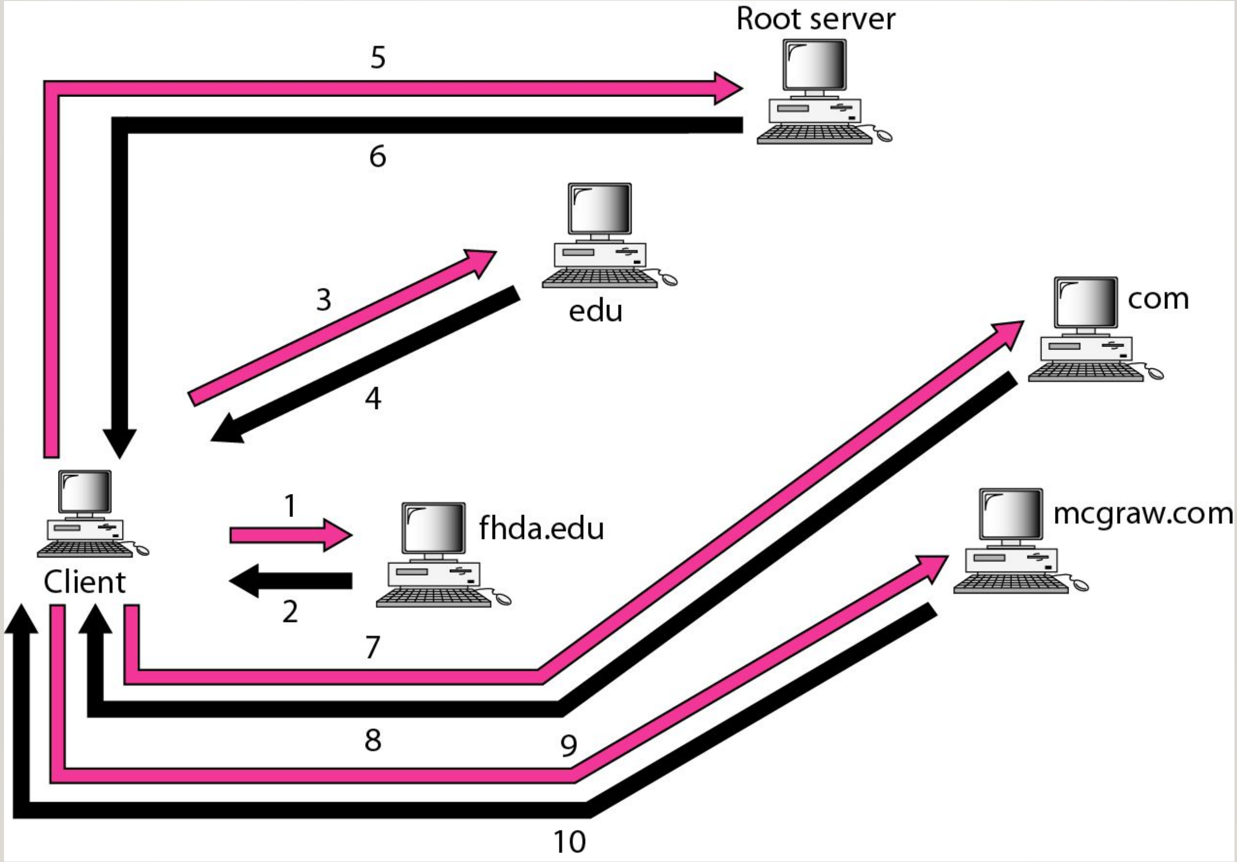
- Mapping an IP address to a domain name



Recursive resolution



Iterative resolution



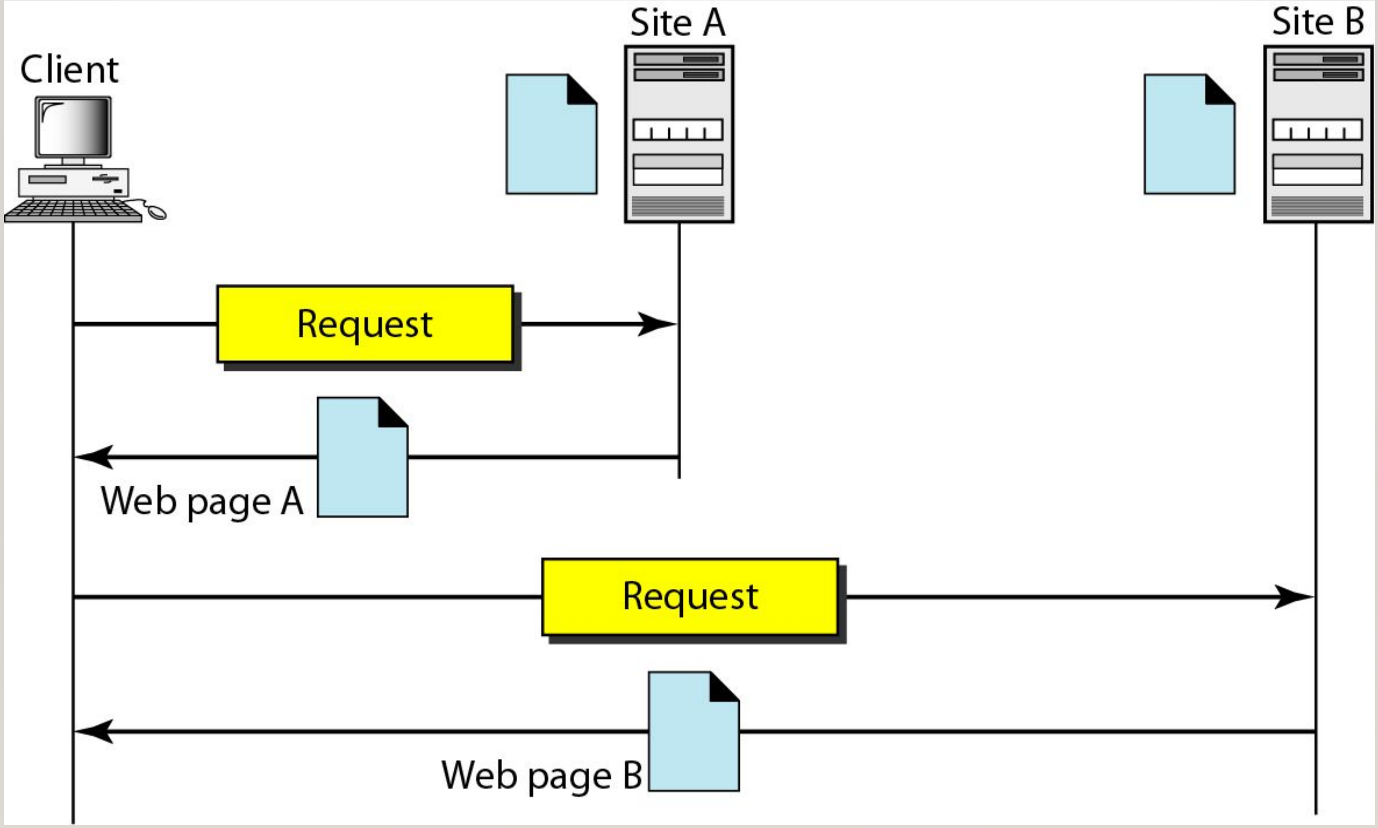
World Wide Web

The WWW today is a distributed client/server service, in which a client using a browser can access a service using a server. However, the service provided is distributed over many locations called sites.

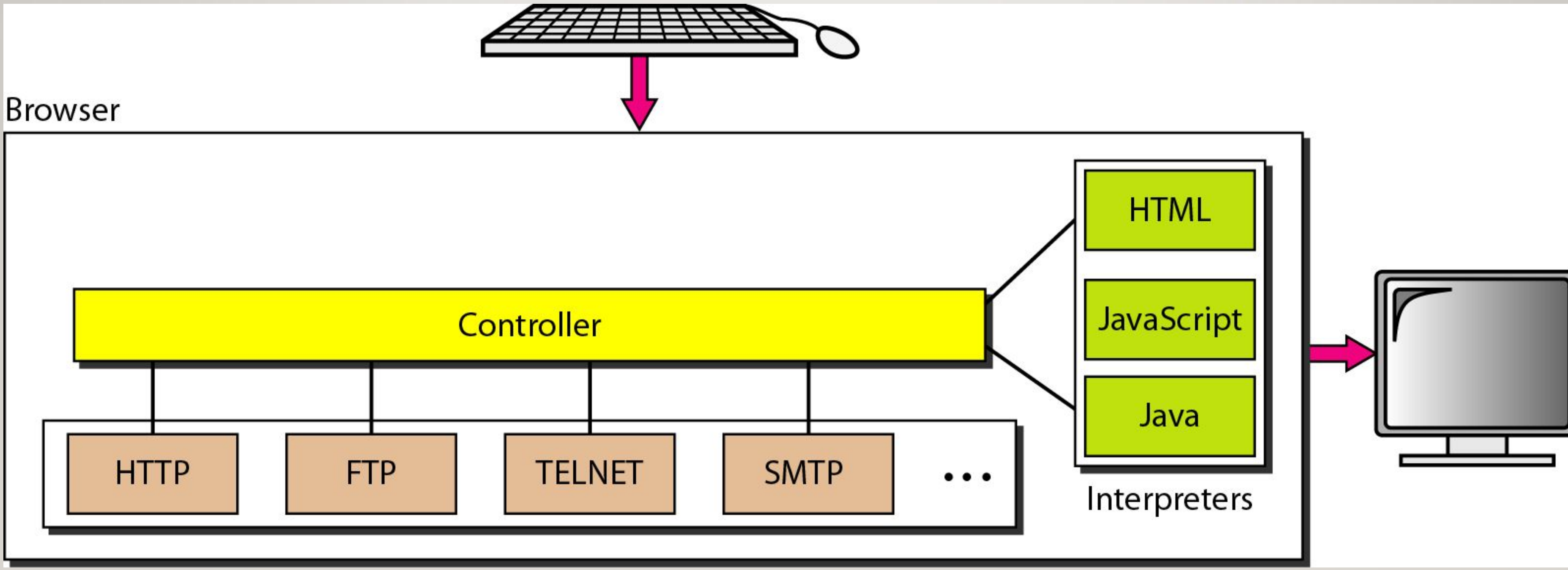
Topics discussed in this section:

Client (Browser)
Server
Uniform Resource Locator

Architecture of WWW



Client Browser



WEB DOCUMENTS

The documents in the WWW can be grouped into three broad categories: static, dynamic, and active. The category is based on the time at which the contents of the document are determined.

Topics discussed in this section:

Static Documents
Dynamic Documents
Active Documents

What is Electronic Mail?

Electronic mail (email) is a method of exchanging messages between people using computers. It's one of the most widely used services on the internet, allowing instant communication across the globe.

Unlike traditional mail, email messages travel through networks at incredible speeds, reaching recipients in seconds rather than days.

Instant Delivery

Messages arrive in seconds, not days

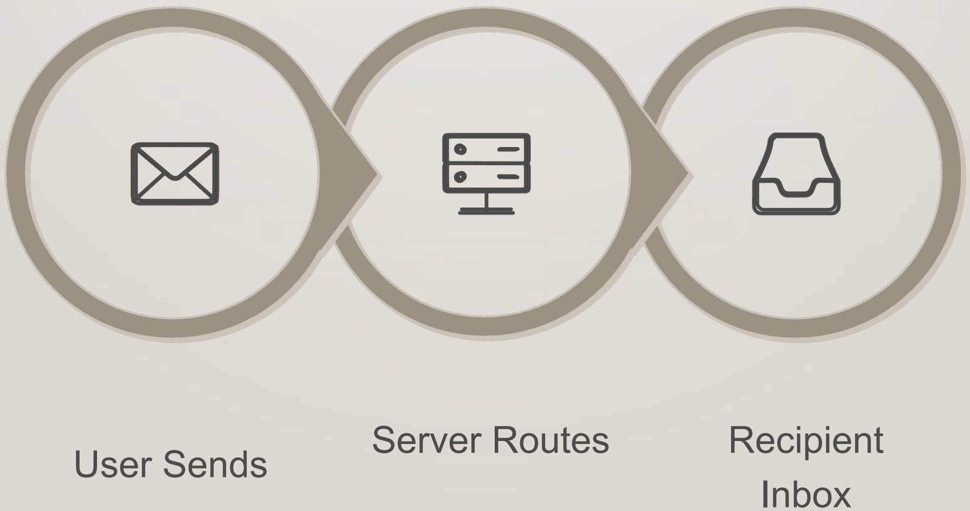
Global Reach

Connect with anyone worldwide

Cost Effective

Free or minimal cost to send

How Email Works: The Basic Process



Mail Servers: The Backbone of Email

Mail Server

A computer that sends, receives, and stores email messages. Think of it as a digital post office that handles your messages.

SMTP Server

Sends outgoing mail using Simple Mail Transfer Protocol. It's responsible for delivering your messages to their destination.

IMAP/POP3 Server

Receives and stores incoming mail. IMAP keeps messages on the server, while POP3 downloads them to your device.

Key Email Components

1	**Email Client** Software you use to write and read emails (Outlook, Gmail, Apple Mail)
2	**Mail User Agent (MUA)** The interface that lets you compose, send, and read messages
3	**Mail Transfer Agent (MTA)** Server software that routes and transfers messages between servers
4	**Mail Delivery Agent (MDA)** Places incoming messages into the recipient's mailbox

Essential Email Protocols

SMTP

Simple Mail Transfer Protocol

Used for sending emails from your client to the mail server and between servers. Port 25 or 587 for secure connections.

POP3

Post Office Protocol

Downloads messages from server to your device. Messages are deleted from server after download. Port 110.

IMAP

Internet Message Access Protocol

Keeps messages on server, allowing access from multiple devices. Synchronizes changes across all devices. Port 143.

MIME

Mail Extension Protocol

Encodes attachments and non-text content for transmission. Enables sending images, documents, and files.

The Complete Email Journey

- Compose
- SMTP Send
- Store
- Deliver & Read

Advantages of Email Systems

Speed

Messages delivered instantly across any distance

Cost-Effective

Minimal cost compared to traditional mail

Global Access

Send messages anywhere in the world

Attachments

Include files, images, and documents

Organization

Sort, search, and archive messages easily

Accessibility

Access from computers, tablets, and phones

Security Issues in Email Systems

Phishing Attacks

Fraudulent emails designed to steal passwords and financial information. Always verify sender addresses before clicking links.

Malware & Viruses

Malicious software attached to emails can infect your computer. Never open attachments from unknown senders.

Spam Messages

Unsolicited bulk emails that clutter inboxes. Use spam filters and don't share your email publicly.

Data Interception

Messages can be intercepted during transmission. Use encrypted connections (SSL/TLS) and avoid sending sensitive data via email.

Key Takeaways

01

Email uses servers and protocols

SMTP for sending, IMAP/POP3 for receiving messages

02

Multiple components work together

Clients, servers, and agents collaborate to deliver messages

03

Fast and cost-effective communication

Instant global reach with minimal cost

04

Security is essential

Protect against phishing, malware, and data interception

Audio streaming

Audio streaming is the continuous transmission of audio data (like music or podcasts) from a server to a client device over the internet. It allows users to listen to media in real time without needing to download files completely to their device storage.

Types of Audio Streaming

On-Demand: Pre-recorded content that users can listen to at any time (e.g., Spotify, Apple Podcasts).

Live: Broadcasting real-time audio over the internet as it is being created (e.g., Internet radio, live concert broadcasts).

Real-Time Interactive: Two-way or multi-way audio communication streams (e.g., VoIP calls, online conferencing).

Audio streaming

Streaming Media Formats

Audio formats refer to the combination of the codec (algorithm compressing/decompressing the data) and the container (the file wrapper holding it).

AAC (Advanced Audio Coding): The default format for most modern streaming services (like Apple Music and YouTube). Offers high fidelity at lower bitrates.

MP3: Highly popular, legacy lossy format. It is universally compatible but removes less noticeable audio frequencies to save bandwidth.

OGG Vorbis: A free, open-source lossy format frequently used by platforms like Spotify.

FLAC (Free Lossless Audio Codec): An audiophile-grade format that compresses audio without losing any original quality, though it requires higher bandwidth.

Audio streaming

Streaming Protocols

Protocols are the underlying network rules dictating how audio data is transmitted, broken into packets, and played back.

- **HLS (HTTP Live Streaming):** Developed by Apple, it segments audio files into small chunks delivered via standard HTTP. It is highly reliable and handles network fluctuations well.
- **WebRTC:** A modern, open-source protocol built for ultra-low latency, peer-to-peer audio communication in web browsers.
- **RTSP & RTP (Real-Time Streaming Protocol/Real-Time Transport Protocol):** Classic protocols used for controlling and transporting real-time audio and video.
- **Icecast & Shoutcast:** Dedicated networking tools specifically designed for broadcasting internet radio stations.

Audio streaming

Different Applications

- **Music Platforms:** Spotify, YouTube Music, Amazon Music.
- **Podcasts:** Apple Podcasts, Google Podcasts.
- **Communication & Conferencing:** Zoom, Microsoft Teams, Discord.
- **Live Broadcasting:** Internet radio stations, Twitch (for live audio/voice channels).

Audio streaming

Advantages

- **Zero Wait Time:** Playback begins almost instantly rather than waiting for massive file downloads.
- **Saves Device Storage:** Requires little to no local memory to store libraries, as data is temporarily cached and cleared via buffering.
- **Convenience & Accessibility:** Provides users instant access to millions of songs and podcasts from any internet-connected device.
- **Live Capabilities:** Enables real-time listening experiences across global audiences.

Disadvantages

- **Internet Dependency:** Offline playback is limited, and buffering/stuttering can occur during poor network conditions.
- **Data Usage:** Constant streaming consumes mobile or broadband data caps.
- **Audio Compression:** Lossy compression formats can slightly reduce audio fidelity compared to uncompressed physical files.

Video streaming

Streaming Media Formats

When video and audio data are processed, they must be compressed and formatted. This relies on two core elements:

- **Codecs:** Algorithms that compress raw media files to save bandwidth without sacrificing significant quality (common examples include \(H.264\) (AVC), \(H.265\) (HEVC), and AV1).
- **Container Formats:** The file types that bundle the compressed audio, video, and metadata together (e.g., MP4, MKV, and WebM).

Applications of Streaming

Streaming technologies power a wide variety of modern tools and platforms:

Entertainment: Video-on-demand services like Netflix and Disney+, and music platforms like Spotify.

Live Broadcasting: Platforms such as Twitch, YouTube Live, and social media streams.

Communication: Video conferencing and unified communications tools like Zoom, Microsoft Teams, and Google Meet.

Security: IP-based surveillance systems and live camera feeds accessed remotely.

Video streaming

Streaming Protocols

Streaming protocols are sets of rules that determine how media packets are sent across networks.

- **HTTP Live Streaming (HLS):** Developed by Apple, this highly popular protocol breaks videos into small, HTTP-based file chunks. It is widely used because it can easily bypass firewalls.
- **Dynamic Adaptive Streaming over HTTP (DASH):** An industry standard that, like HLS, breaks video into smaller segments. It is designed to be codec-agnostic.
- **Real-Time Messaging Protocol (RTMP):** A legacy protocol traditionally used for streaming from an encoder to a server.
- **Real-Time Streaming Protocol (RTSP):** Used frequently in IP security cameras and remote production environments to control the media server. To ensure a flawless viewing experience, both HLS and DASH employ **Adaptive Bitrate Streaming (ABR)**. ABR constantly assesses the viewer's internet bandwidth and device processor capabilities, dynamically adjusting the video quality (e.g., swapping between 1080p, 720p, and 480p) to prevent buffering.

Video streaming

Advantages

- **Instant Access:** Viewers do not have to wait for large files to finish downloading before playing the content.
- **Low Storage Requirements:** Content exists in the cloud, meaning client devices require minimal local storage.
- **Copyright Protection:** Because the data is delivered dynamically, users do not own the raw file, which helps prevent piracy.

Disadvantages

- **Internet Dependent:** Playback requires an active, stable internet connection.
- **Buffering & Lag:** Network congestion or dropped packets can cause latency, interruptions, or a sudden drop in visual quality.
- **Server/Network Demands:** Live streaming and concurrent high-traffic events demand rigorous backend server and network infrastructures to operate smoothly